Spatial Changes in Sea Ice Fragmentation and Configuration in the Weddell Sea Using GIS Techniques

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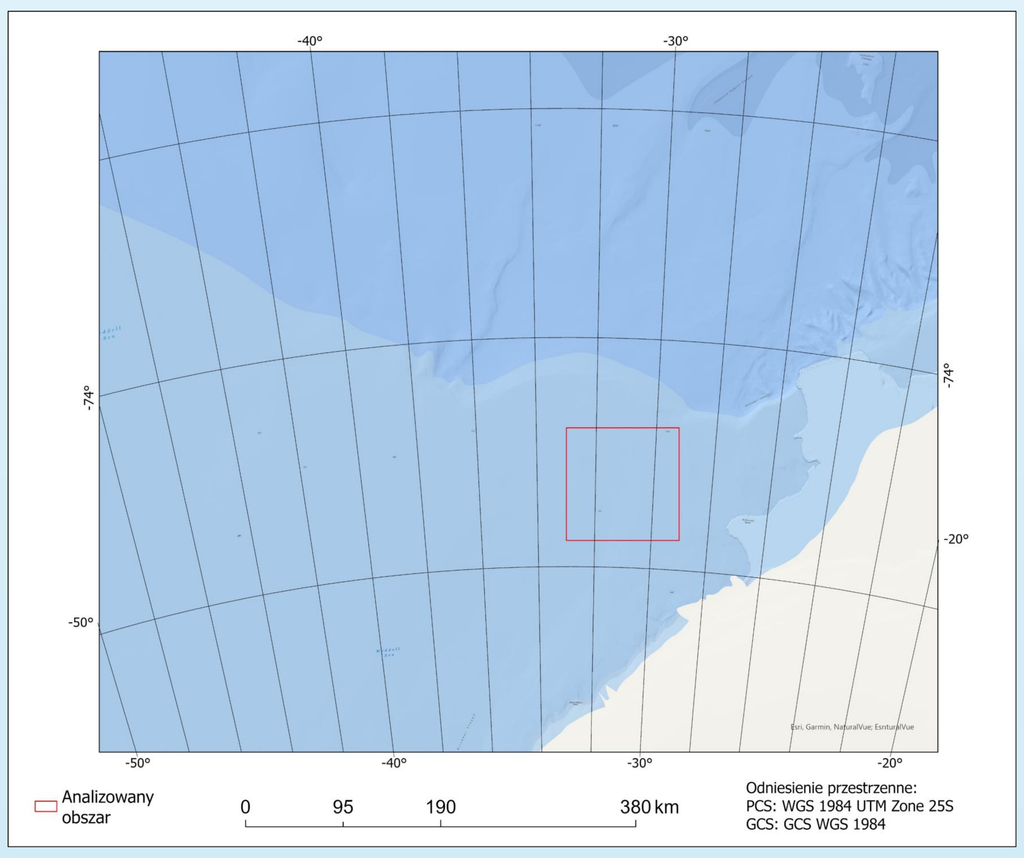
BACKGROUND

Why the Weddell Sea?

- A key component of the global climate system through its role in ocean circulation and climate regulation.
- One of Antarctica's most important biodiversity hotspots.



Research Questions:

- Did the total sea ice extent within the study area differ between January 2019 and January 2021?
- Were there significant differences in the spatial fragmentation of sea ice between the two observation periods?
- To what extent can GIS techniques applied to Sentinel-2 imagery be used to identify and quantify spatial changes in sea ice cover?

Objective:

- The objective of this study was to employ Sentinel-2 satellite imagery and GIS techniques to identify, extract, and analyze the spatial characteristics of sea ice in the Weddell Sea, and to quantify changes in both ice extent and fragmentation between January 2019 and January 2021.

METHODOLOGY

Data Processing Workflow

Acquisition of Sentinel-2 imagery from the Copernicus Data Space Ecosystem and import of satellite data into ArcGIS Pro.

Reflectance correction for bands B2, B3, B4 and B11

$$Ref = \frac{Band}{10000}$$

Generation of an RGB composite (B4–B3–B2).

Calculation of the Normalized Difference Snow Index (NDSI):

$$NDSI = \frac{B3\ - B11}{B3 + B11}$$

where B3 – Green band and B11 – Short-Wave Infrared (SWIR) band

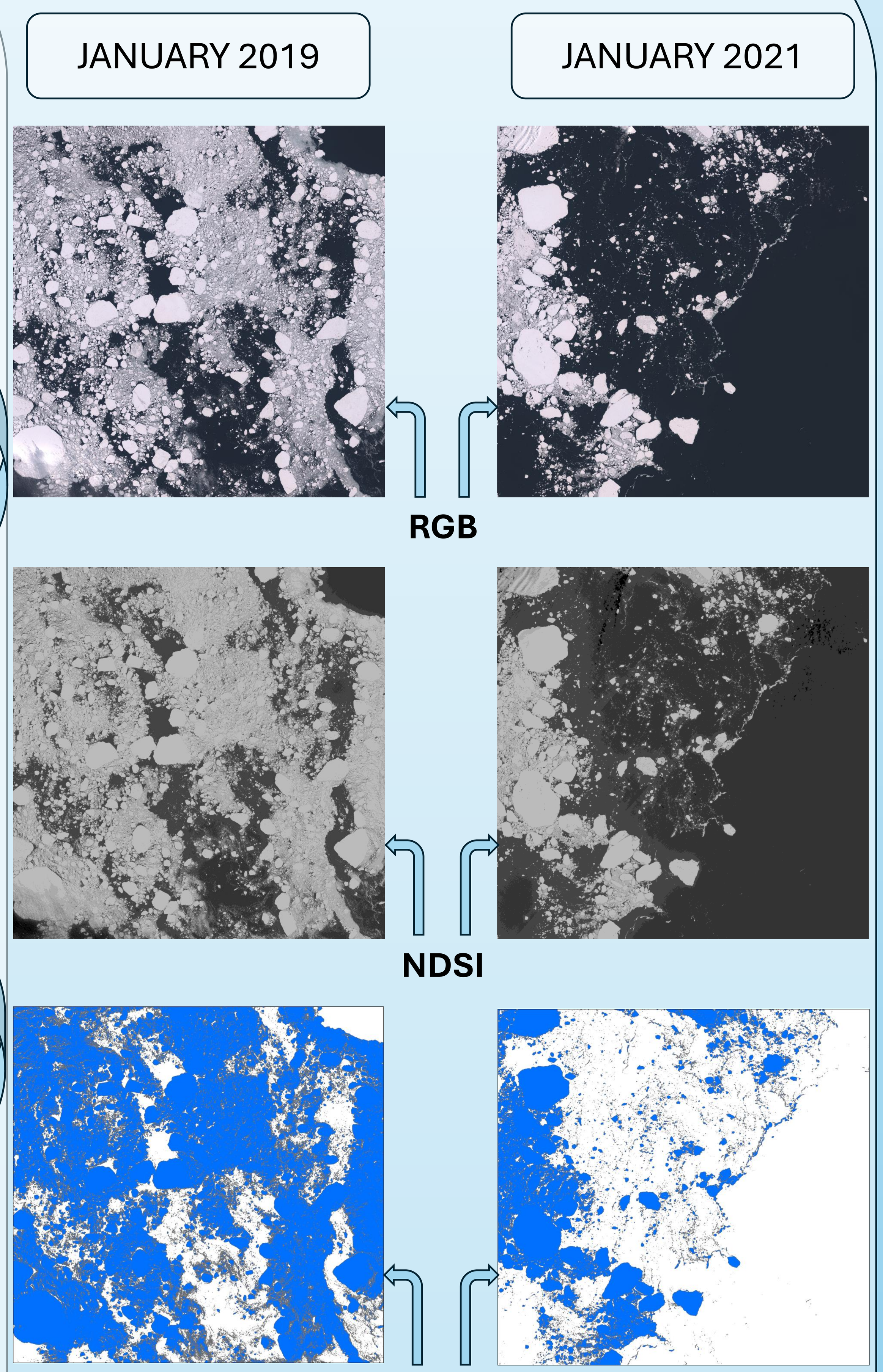
Majority filtering to reduce classification noise.

Classification of sea ice and open water.

Polygonization of the extracted sea ice cover.

Spatial analysis of ice-cover characteristics.

RESULTS



Polygonization

Parameter	January 2019	January 2021
Study area (km^2)	12,056.02	12,056.02
Number of ice floes	48,584	20,753
Total sea ice area (km^2)	8,379.91	2,669.17
Mean ice floe area (km^2)	0.17	0.13
Ice floe density	4.03	1.72
Sea ice concentration	69.51%	22.14%

Conclusions:

- Total sea ice extent decreased from 8,379.91 km^2 to 2,669.17 km^2, corresponding to a reduction in sea ice concentration from 69.51% to 22.14% within the study area.
- The number of detected ice floes decreased by approximately 57%, indicating substantial changes in the spatial organization of the sea ice cover.
- The mean area of individual ice floes remained relatively stable, suggesting that the dominant process was a reduction in the number of ice floes rather than significant changes in their average size.
- Sentinel-2 imagery combined with GIS-based image analysis proved to be an effective approach for detecting and quantifying changes in sea ice extent and spatial configuration.

Limitations and Future Work:

- (1) The analysis is based on only two observation dates and therefore does not capture long-term trends in sea ice variability. (2) The study is limited to a single region of the Weddell Sea, so the findings should not be generalized to the entire Antarctic sea ice system.
- (1) Extend the analysis to multi-year time series in order to investigate seasonal and long-term variability. (2) Develop automated methods for iceberg detection and monitoring using consecutive Sentinel-2 observations.

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